MILITARY SPECIFICATION

SYNCHRO BOOSTER AMPLIFIER MK 162 MOD 0

This specification has been approved by the Naval Ordnance System Command Headquarters, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of the Synchro Booster Amplifier Mark 162 Mod 0. This amplifier is of one class and one type.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood
PPP-B-621	Boxes, Wood
PPP-B-636	Boxes, Fiber

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-S-901	Shockproof Equipment, Class HI (High Impact) Shipboard Application, Tests for
MIL-E-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts

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MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use; the General Specification for Design and Procurement of
MIL-W-21927	Weapons, Handling and Preparation for Delivery of; General Requirements for

STANDARDS

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 162316	Synchro Booster Amplifier Mk 162 Mod 0, and All Parts and Sub-assemblies Listed Therein

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

WR-11	Design and Test of Packaging, Packing, Shipping and Handling Equipment for Weapon System Components
QAP	Quality Assurance Provisions for Synchro Booster Amplifier Mk 162 Mod 0

(Copies of specifications, standards, and drawings requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois. Orders for the above publication should cite the latest edition and supplements thereto.)

AMERICAN TRUCKING ASSOCIATION, INC. PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street N.W., Washington 6, D.C.)

3. REQUIREMENTS

3.1 Description. - The synchro booster amplifier is a solid-state amplifier designed to increase the power level outputs of 60 or 400 cps synchro circuits. A self-contained power supply develops DC voltages for operation of two identical channels, each capable of amplifying the power output of a synchro transmitter. Increased power output permits larger load carrying capabilities and an increase in the number of synchro receivers which may be connected to an existing circuit.

The amplifier is housed in a 10-inch square, cast aluminum housing, and weighs 60 pounds.

3.2 General Requirements

3.2.1 First Article Samples. - Unless otherwise specified in the contract or order, first article samples of the amplifier shall be manufactured using the methods and procedures proposed for the production lot. The sample shall be tested as per Sect. 4 herein and is for the purpose of determining that the contractor's methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the amplifier shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications and publications listed in BUWEPS LD 162316.

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3.2.3 Finishes, Coatings, Construction and Dimensions. - Finishes, coatings, construction and dimensions for the amplifier shall be strictly in accordance with the applicable drawings, specifications and publications listed in BUWEPS LD 162316.

3.2.4 Moisture, Fungus and Corrosion. - The amplifier shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in BUWEPS LD 162316.

3.2.5 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in BUWEPS LD 162316.

3.2.6 Interchangeability - Individual parts of assemblies, fabricated in agreement with their applicable drawings and specifications shall be interchangeable with mating parts of the unit and other like units. If deviation greater or smaller than the tolerances shown on the drawings or specifications is required for interchangeability, or if hand fitting beyond that shown on applicable drawings is required, then changes in dimensions are necessary and revision and correction of the applicable drawings are required. (See 3.2.9.)

3.2.7 Threaded parts - Screw threads shall conform to MIL-S-7742.

3.2.8 Requests for Deviation. - Each request that a deviation be authorized shall have been submitted to the procuring activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the procuring activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.2.9 Conflicting Requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

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3.2.10 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy final inspection gages for guidance in the design of the contractor's inspection gages. However, such procedures shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy final inspection gages.

3.3 Performance Requirements and Product Characteristics. - The amplifier shall meet the following performance requirements and product characteristics:

3.3.1 Dielectric Strength. - The amplifier shall be capable of withstanding a potential of 300 $\pm \frac{0}{20}$ volts DC for a period of one minute, without evidence of dielectric failure.

3.3.2 Insulation Resistance. - The amplifier shall show an insulation resistance of at least 3 megohms when subjected to a potential of 500 volts DC.

3.3.3 Power Supply, Operating Voltages. - The power supply outputs shall be within the limits listed in Table 1 when tested as defined in Section 4.

Table 1

Test Points	Limits
A3A1 - TP-A (+) to TP-C (-)	74 ± 8.0 VDC
A3A1 - TP-B (-) to TP-C (+)	74 ± 8.0 VDC
A3A1 - TP-D (+) to TP-C (-)	29 ± 2.4 VDC
A3A1 - eyelet 15 (+) to TP-C (-)	60 ± 6.0 VDC

3.3.4 Voltage Gain, 60 cps or 400 cps. - Both amplifier channels shall be capable of producing a minimum open loop voltage gain of 100 to 1.

3.3.5 Linearity. - Over an input voltage range of 1 volt to 90 volts, the response generated within the amplifier shall be directly proportional to the input voltage.

3.3.6 Power System Variations. - The amplifier shall function in accordance with the performance requirements of this specification during the power variation test specified in 4.4.8.

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3.4 Environmental Requirements

3.4.1 Watertightness. - The amplifier shall show no entry of water and shall meet the performance and other requirements as specified in this specification after being subjected to the watertight test specified in 4.4.3.

3.4.2 Inclination. - The amplifier shall be capable of withstanding inclination tests as specified in 4.4.4, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.3 Vibration

3.4.3.1 Transportation. - The amplifier shall be capable of withstanding the vibration tests specified in 4.4.5.1 and thereafter show no damage and shall meet all the requirements of Section 3.

3.4.3.2 Operating. - The amplifier, less pack and packaging, shall be capable of withstanding the vibration tests specified in 4.4.5.2. During and after the tests, the amplifier shall show no damage and shall meet all the performance requirements and product characteristics of this specifiaction.

3.4.4 Shock

3.4.4.1 Transportation, Storage and Handling. - The amplifier shall be capable of withstanding the shock tests specified in 4.4.6.1 and thereafter shall show no damage and shall meet all the requirements of Section 3.

3.4.4.2 Operating. - The amplifier, mounted as specified in 4.4.6.2, shall be capable of withstanding shock tests as specified in 4.4.6.2. During and after the test, the amplifier shall show no damage and shall meet all performance requirements and product cahracteristics of this specification.

3.4.5 Temperature

3.4.5.1 Storage. - The amplifier shall be capable of being stored without damage in any ambient temperature from minus 62 degrees C. to plus 75 degrees C.

3.4.5.2 Operating. - The amplifier shall be capable of satisfactory operation in any ambient temperature from 0 degrees C. to plus 50 degrees C.

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3.4.6 Humidity. - The amplifier shall be capable of withstanding humidity tests specified in 4.4.9, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.5 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. - The amplifier including all parts and accessories shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved first article sample, subject to any qualification stated in the Governments notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspections. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests

4.2.1 Quality Conformance Inspections

4.2.1.1 Individual. - Unless otherwise specified, all units submitted for acceptance shall perform the following tests. Failure of any unit to meet acceptance criteria for all tests shall be cause for rejection of that unit.

Visual (4.4.1)
Tests in accordance with QAP Document (4.4.2)

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4.2.1.2 Sampling. - When so specified in the contract or order, the following tests of Sampling Plan A or Sampling B shall be performed on a sample group of Amplifier Mark 162 Mod 0 units representative of the normal production of the manufacturer. Sampling inspections shall be performed prior to submission of each lot. Failure of any sample unit to meet the requirements of 4.2.1.1 after any Sampling test shall be cause for rejection of the lot. Sample size and lot size shall be as specified by the procuring activity. Provision for rework or disposition of samples shall be as specified in contract or order. (see 6.2)

Sampling Plan A
- Vibration (4.4.5)
- Inclination (4.4.4)
- Degree of enclosure (4.4.3)
- Power (4.4.8)

Sampling Plan B
- Rough handling (4.4.6)
- Shock (4.4.6)
- Temperature (4.4.7)
- Humidity (4.4.9)
- Power (4.4.8)

4.3 Test Conditions

4.3.1 Environmental Conditions. - Unless otherwise specified all tests shall be conducted under the following ambient conditions:

Temperature - Room ambient (65° F to 110° F)
Atmospheric pressure - Room ambient
Humidity - Room ambient (95 per cent relative, max.)
Vibration - Room ambient

4.3.2 Test Equipment. - The test equipment used shall be of sufficient accuracy to measure the absolute values of the required variables within ten percent of their respective tolerances.

4.4 Test Methods

4.4.1 Visual. - Each amplifier shall be visually examined to assure that the assembly and all parts comply with 3.5 and 3.6. Evidence of sub-standard workmanship shall be cause for rejection.

4.4.2 Tests in Accordance With QAP Document. - Tests shall be performed on each amplifier, in accordance with the QAP document, to determine compliance with the appropriate requirements of this specification. Failure of any amplifier to comply with all requirements of the QAP document shall be cause for rejection of that unit.

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4.4.3 Watertightness. - The amplifier shall be tested in accordance with the watertight tests specified in Standard MIL-STD-108.

4.4.4 Inclination. - The amplifier shall be tested for satisfactory operation while undergoing the inclination tests outlined in Specification MIL-F-18870. At the conclusion of the tests, the amplifier shall meet the performance and other requirements specified in this specification.

4.4.5 Vibration

4.4.5.1 Vibration, Transportation. - The amplifier, packaged and packed as specified in Section 5, shall be vibration tested in accordance with procedures specified in Weapons Requirements WR-11.

4.4.5.2 Operating. - The amplifier shall be subjected to the vibration tests specified in Specification MIL-F-18870.

4.4.6 Shock

4.4.6.1 Shock, Transportation, Storage, and Handling. - The amplifier, packed and packaged as defined in Section 5, shall be subjected to the appropriate shock isolation tests specified in Weapons Requirements WR-11.

4.4.6.2 Operating. - The amplifier shall be mounted on the standard mounting bracket of a shock testing machine in a manner most closely simulating actual service installation. The switch box shall be energized to perform its normal function in the operating condition, and shall be subjected to shock tests specified in Specification MIL-S-901.

4.4.7 Temperature

4.4.7.1 Storage. - The amplifier shall be subjected to the temperature tests specified in Specification MIL-F-18870, Non-operating Conditions, Class 4c.

4.4.7.2 Operating. - The amplifier shall be subjected to the temperature tests specified in Specification MIL-F-18870, Operating Conditions, Class 4c.

4.4.8 Power System Variations. - The amplifier shall be tested for satisfactory operation under power system variations as specified for line surges in Specification MIL-F-18870. This test may be run during the performance of any or all of the tests given in 4.4.3 to 4.4.7 inclusive of this specification at the direction of the testing agency, or as directed by the procuring activity.

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4.4.9 Humidity. - The amplifier shall be subjected to the humidity tests specified in Specification MIL-F-18870.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, Drying and Preservation Application. - Not applicable.

5.1.1.2 Unit Packaging. - Unit packaging of the amplifier shall be in accordance with Method IIb of Specification MIL-P-116. The unit package shall contain one amplifier packaged in fiberboard boxes in accordance with Specification PPP-B-636 Class 2.

5.1.1.3 Intermediate Package. - Not applicable.

5.1.1.4 Cushioning. - The unit package, and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the amplifier will meet all the requirements of Section 3 after being tested in accordance with 4.4.5.1 and 4.4.6.1.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, Drying and Preservative Application. - Not applicable.

5.1.3.2 Unit Packaging. - Unit packaging of the amplifier shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one amplifier.

5.1.3.3 Intermediate Package. - Not applicable.

5.2 Packing

5.2.1 Level A. - In addition to the following, Level A packing shall be in accordance with Specification MIL-P-17555.

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5.2.1.1 Exterior Containers. - One (1) amplifier packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-621 Class 2, or PPP-B-601 (overseas).

5.2.2 Level B. - In addition to the following, Level B packing shall be in accordance with Specification MIL-P-17555.

5.2.2.1 Exterior Containers. - One (1) amplifier packaged as described in 5.1.1.2 shall be packed in a fiberboard box in accordance with Specification PPP-B-636, Class 1.

5.2.3 Level C. - The amplifier packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special Marking. - None, unless otherwise specified.

5.3.2 Normal Marking. - In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended Use. - The Synchro Booster Amplifier Mark 162 Mod 0, is intended for use in Fire Control Systems Mark 105 Mods 23 and 27, to increase the efficiency of Relay Transmitter Mark 56 Mod 1. The amplifier may also be used in other applications where it is desired to increase synchro signal power.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Sampling tests required (4.2.1.2)

c. Sample size (4.2.1.2)

d. Lot size (4.2.1.2)

e. Disposition of test samples (4.2.1.2)

f. Level of "Preparation for Delivery" (see Section 5).

g. Additional marking if required (see 5.3.1).

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6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to paragraph 4.2.1 of this specification.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for six months or less or in domestic locations for an indefinite period.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any amplifier which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

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DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-82464(OS) SYNCHRO BOOSTER AMPLIFIER MK 162 Mod 0

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *IF "YES", IN WHAT WAY?*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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